

6 ai $E_p = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(570 \times 10^{-9})}$ **M1**

$E_p = 3.5 \times 10^{-19} \text{ J}$ **A0**

aii Number of photons in 1.0 s,

$P = NE / t$

$N / t = (2.7 \times 10^{-3}) / (3.5 \times 10^{-19})$

$= 7.714 \times 10^{15}$ **M1**

Using de Broglie's equation, initial momentum p is

$p = \frac{h}{\lambda}$

$p = \frac{(6.63 \times 10^{-34})}{570 \times 10^{-9}}$ **M1**

$p = 1.163 \times 10^{-27} \text{ N s}$

The final momentum of each photon is zero due to no reflection of light.

Total change in momentum of photons = $N \Delta p$ **C1**

$= |(7.714 \times 10^{15})(0 - 1.163 \times 10^{-27})|$

$= 8.97 \times 10^{-12} \text{ kg m s}^{-1}$. **A1**

bi Using Fig. 6.2, threshold frequency $f_0 = 1.0 \times 10^{15} \text{ Hz}$ ($\pm 0.1 \times 10^{15} \text{ Hz}$). **C1**

work function energy = hf_0

$= (6.63 \times 10^{-34}) (1.0 \times 10^{15})$

$$= 6.63 \times 10^{-19} \text{ J} \quad \text{C1}$$

$$= 4.14 \text{ eV} \quad \text{A1}$$

*max of 2 marks if read wrongly (process must be right)

- bii**
- Straight line with the same gradient and
 - Displaced to the right
- A1**

- biii** Intensity determines the rate of photons arriving on the metal (or the rate of emission of the electrons from the metal). **B1**

Intensity does not affect the energy of the photons, **B1**

Hence the graphs in (i) and (ii) do not depend on the intensity of the incident radiation **A0**

- ci** distance = speed $\times$ time
- $$2.6 \times 10^{-10} = 3.00 \times 10^8 \times \text{time}$$
- $$\text{time} = 8.67 \times 10^{-19} \text{ s} \quad \text{A1}$$

- cii** maximum uncertainty in time = time taken to travel atom
- $$= 9 \times 10^{-19} \text{ s (1 s.f.)} \quad \text{A1}$$

- ciii** $8.67 \times 10^{-19} \times \Delta E \geq 6.63 \times 10^{-34}$
- $$\Delta E \geq 7.65 \times 10^{-16} \text{ J}$$
- $$\Delta E \geq 8 \times 10^{-16} \text{ J} \quad \text{C1}$$

Uncertainty of energy of photon is larger than the energy of photon ($3.5 \times 10^{-19} \text{ J}$) **C1**

Hence the energy of the photon may not be the same. **A0**